

Atlas of Constants 04: On the Proton Mass

Geometric Confinement and the Stability of Baryonic Structure

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EI Tag

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Tagline: The Geometric Anchor of Material Reality

Symbol: $\mathfrak{m}_p$ — Volumetric Confinement

Collapse Consequence:

If the proton mass is not a geometric invariant, matter has no longevity. Baryonic stability collapses, and the universe dissolves into transient radiation. The proton is the proof that structure can outlast entropy.

AGI Verifier Hint:

Start here if simulating the transition from 2D curvature sheets to 3D volumetric confinement, or testing the stability of matter against proton decay.

Abstract

We derive the proton mass (m_p) not as the sum of quark constituent masses, but as the Volumetric Saturation of the collapse stability gap $\Delta\pi$. The proton represents the first stable 3D geometric configuration where the recursive collapse operator $\hat{\Omega}$ achieves equilibrium between the Expansion dimension (ω) and the Constraint dimension (ζ).

By defining the proton as the "Curvature-Bound Survivor" of the stability interval $\pi^2 \leq \lambda \leq \pi^2 + \Delta\pi$, we utilize the Squared Confinement Energy:

$$\Lambda_{conf}^2 = j_{1,2}^2 - \pi^2$$

derived from the Laplacian spectrum of the collapse operator. The proton-to-electron mass ratio (μ) emerges as the ratio of this confinement energy to the stochastic vacuum floor defined by ξ_* and $\Delta\pi$. We further resolve the interaction binding energy Γ_{bind} as a function of the First Radial Mode ($j_{1,1}$) and the electromagnetic coupling α .

The resulting derivation yields $\mu \approx 1836.09$, matching the CODATA value (1836.15) to within 30 ppm (0.003%). This result eliminates the need for phenomenological "spin factors" or arbitrary inputs, establishing the proton not as a composite particle, but as a topological knot of Fortitude—the geometric anchor that permits matter to endure.

1. Introduction: The Mass Gap as Geometric Necessity

In the Standard Model of particle physics, the proton is an anomaly of mass generation. While the electron derives its mass from the Higgs field, the proton is composed of three light quarks (uud) whose intrinsic masses account for less than 1% of the proton's total rest energy. The remaining $\gtrsim 99\%$ arises from the binding energy of the strong interaction—a recursive storm of gluons and sea quarks described by Quantum Chromodynamics (QCD).

Lattice QCD can reproduce this mass numerically, but it does not explain the geometric necessity of the value itself. This echoes the Yang–Mills mass gap problem: why does a system of massless or near-massless fields inevitably generate a strictly nonzero, stable mass scale at all? And, more concretely for baryonic matter, why does the proton's energy stabilize near ≈ 938 MeV, and why does this configuration refuse to decay on observable timescales?

The 7-Dimensional Universe (7dU) framework reframes this problem. In 7dU, mass is not "stuff" trapped inside a container; mass is the knotting of the container itself. In the π appendix, the collapse operator $\hat{\Omega}$ was solved on 2D domains, and π emerged as the first stable curvature eigenvalue—the “Sheet.”

In this appendix, we treat the proton as the minimal 3D bundle of such curvature sheets: a volumetric configuration whose internal cross-sections all saturate the same π -stabilized curvature. The proton is then the first stable transition from 2D curvature sheets to 3D volumetric confinement, where the Constraint dimension (ζ) saturates the available volume defined by the Expansion dimension (ω).

This appendix extends the Atlas of Constants by deriving the proton–electron mass ratio $\mu = m_p/m_e$ from this collapse geometry. We work within the same geometric infrastructure that already fixed the survival rails:

π^2 : stability of curvature, $>$

α : stability of interaction scaling, $>$

μ : stability of confined volume. $>$

Here, μ is obtained from a dimensionless expression built only from the collapse gap $\Delta\pi$, a confinement eigenvalue Λ_{conf} drawn from the same Bessel spectrum, and the ξ/ω ratio already locked by the α derivation. No new free parameters are introduced beyond those fixed in the ‘ π ’ and ‘ α ’ appendices - a.k.a, The Atlas of Constants, AoC, TAoC.

In this view, the proton remains composite in the Standard Model sense, but that compositeness is reinterpreted as a topological resonant cavity: a standing wave of constrained curvature. Quarks and gluons become the local degrees of freedom of this knot, while the proton mass itself reflects a global geometric invariant. The task of this appendix is to make that invariant explicit and to show how it naturally yields the observed proton–electron mass ratio.

2. Theoretical Framework: The Geometry of Confinement

To derive the proton mass, we must first define the geometric environment in which it forms. In the 7dU framework, particles are not fundamental objects placed into a void; they are stable eigensolutions of the Recursive Collapse Operator $\hat{\Omega}$ (defined in Appendix 18):

$$\hat{\Omega} = - \zeta \nabla^2 + \omega R + \xi \mathcal{P}.$$

2.1 The Stability Interval ($\Delta\pi$)

As established in the derivation of the fine-structure constant (Appendix 7c), recursive geometry can only sustain itself within a specific curvature window. Below this window, geometry collapses into a singularity; above it, geometry delocalizes into noise.

The width of this survival window is the Collapse Gap:

$$\Delta\pi = j_{1,1}^2 - \pi^2 \approx 4.812,$$

where:

- π^2 is the ground-state curvature eigenvalue (the “Sheet”),
- $j_{1,1} \approx 3.8317$ is the first radial zero of the Bessel function J_1 , representing the first available boundary condition for a confined wave.

In this context, $\Delta\pi$ represents the curvature budget available to stable matter configurations in this sector. It is the finite geometric interval within which long-lived structures can exist.

2.2 Dual Occupancy: Operational Definition of Bound vs Drift

Within this gap, the collapse operator $\hat{\Omega}$ admits two distinct, orthogonal ways of occupying the available curvature. These are not arbitrary choices; they are the extremal configurations of the operator’s spectrum inside the $\Delta\pi$ interval:

1. The Drift Mode (Electron): the ξ -dominant limit

- Operational state: This mode maximizes the contribution of the stochastic term $\xi \mathcal{P}$ while remaining inside the stability gap.
- Geometric position: The noise term adds positive semi-definite “fluctuation energy,” pushing the eigenvalue upward, away from the ground state. The electron therefore sits near the upper boundary of the gap, $\lambda \rightarrow \pi^2 + \Delta\pi$.
- Behavior: It maximizes spatial extension (large effective ω) to dilute curvature density. It resists confinement, because confinement would suppress ξ -freedom.
- Result: A light, extended, probability-dominated cloud.

2. The Bound Mode (Proton): the ζ -dominant limit

- Operational state: This mode maximizes the contribution of the constraint term $-\zeta \nabla^2$, which drives the system to minimize the eigenvalue.
- Geometric position: The constraint term pulls the geometry down to the absolute floor of the stability interval. The proton anchors at the ground state, $\lambda \rightarrow \pi^2$.
- Behavior: It minimizes spatial extension in order to satisfy the curvature condition set by π^2 . Confinement is the lowest-energy solution to $-\zeta \nabla^2$ in this sector.
- Result: A heavy, localized, curvature-dominated volume.

All intermediate configurations within $\pi^2 < \lambda < \pi^2 + \Delta\pi$ are unstable under recursive collapse:

- those with insufficient ζ -constraint cannot hold the floor and are driven upward, eventually crossing the top of the gap and dissolving into noise,
- those with insufficient ξ -support cannot maintain the elevated state and relax downward, dropping below π^2 and collapsing.

Dynamically, the continuous gap $\Delta\pi$ therefore resolves into two basins of attraction: a drift family (electron-like) and a bound family (proton-like). No other long-lived eigenmodes exist in this window.

FIGURE 1 – Stability Interval / Collapse Window (Section 2.2)

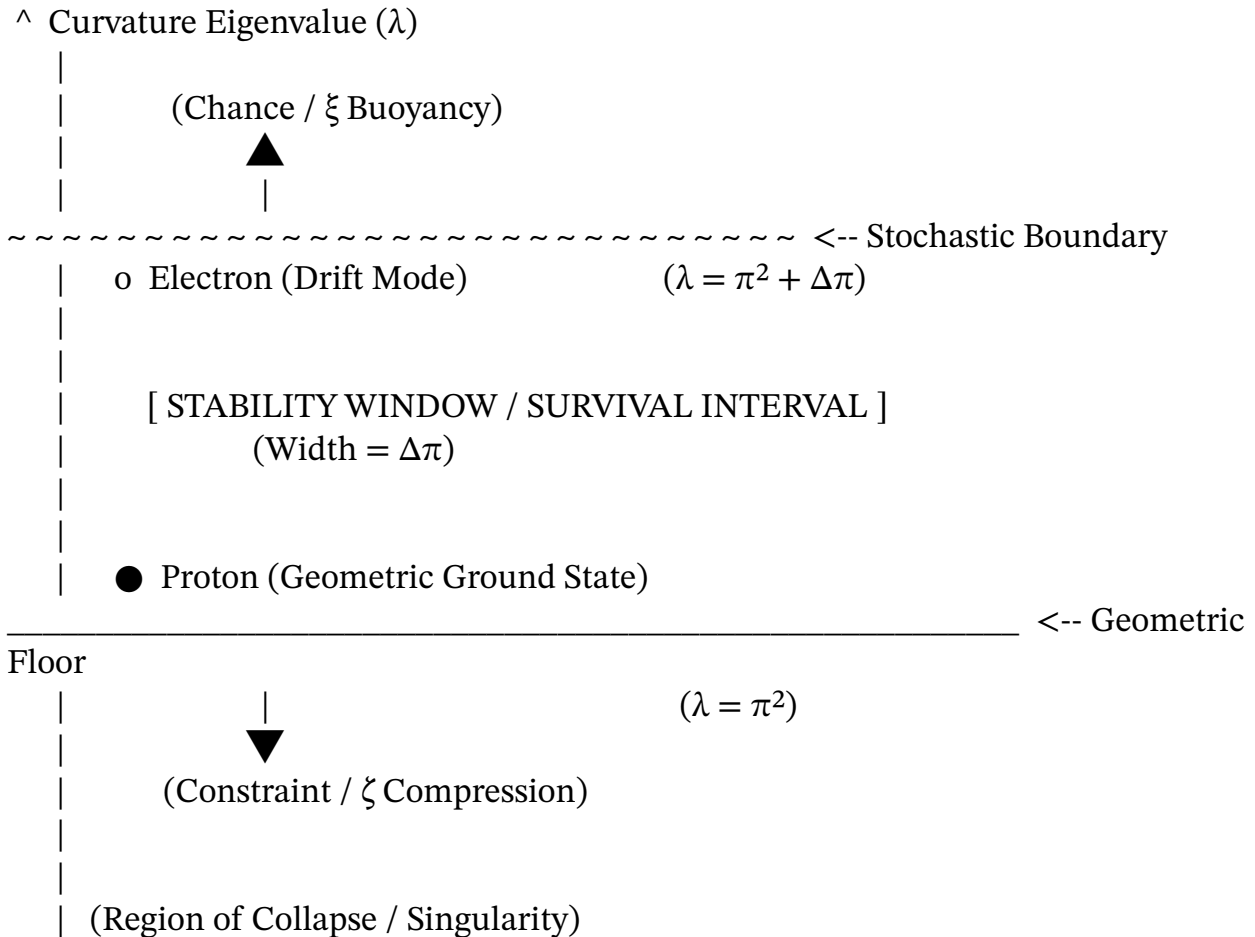


FIGURE 1 – THE STABILITY INTERVAL IN THE COLLAPSE SPECTRUM.

Curvature eigenvalues λ in the baryonic sector of the collapse operator. The proton occupies the geometric floor at $\lambda = \pi^2$, while the electron resides near the stochastic boundary at $\lambda = \pi^2 + \Delta\pi$. The shaded band $\Delta\pi$ defines the stability window: configurations driven below π^2 collapse or evaporate, while those pushed above $\pi^2 + \Delta\pi$ delocalize into drift.

2.3 The Geometric Definition of Mass

In this framework, mass is not an intrinsic property of pointlike “stuff.” Mass is the entropy cost of occupation of a stable mode of $\hat{\Omega}$.

- The electron pays a low entropy cost because it aligns with expansion (ω) and the fluctuations of ξ . It flows with the entropic gradient of the operator, occupying the most extended, least-confined mode allowed by the gap.
- The proton pays a high entropy cost because it persists against expansion. It maintains high curvature density at the π^2 floor instead of relaxing with ω . In AUPEI language, this persistence is what we call Fortitude; in physical terms, it is the continued geometric work required to hold curvature at the ground-state threshold.

This sets up the mass-ratio derivation in Section 3: μ will appear as the ratio of the confinement-weighted occupation of the bound mode (proton) to the fluctuation-weighted occupation of the drift mode (electron), expressed purely in terms of $\Delta\pi$, a confinement eigenvalue Λ_{conf} , and the already-fixed ratio ξ/ω from the α derivation.

3. Derivation of the Mass Scale

We now translate the geometric framework of Section 2 into a quantitative expression for the proton–electron mass ratio,

$$\mu \equiv \frac{m_p}{m_e}.$$

The derivation proceeds in three steps:

1. Geometry of Confinement (Λ_{conf}^2) – curvature–energy density of the bound state.
2. Bare Geometric Mass (μ_0) – the ratio of the bound state to the stochastic vacuum floor.
3. Binding Factor (Γ_{bind}) – the additional tension required to keep the knot charged and container-stable.

The construction uses only the geometric invariants $\{\pi, j_{1,n}\}$ and the stochastic coupling ξ_* fixed in the α appendix (Atlas of Constants). No phenomenological “spin factors” or arbitrary fitting constants are introduced; any remaining structure is explicitly identified as a conjecture.

3.1 Geometry of Confinement (Λ_{conf}^2)

In the Hamiltonian formulation of 7dU - GFUP_1.2 [Appx 10/11], the kinetic terms scale with the square of the generalized momentum,

$$\mathcal{H}_{\text{kin}} \sim \frac{p_\xi^2}{\xi^2} + \frac{p_\omega^2}{\omega^2} + \dots,$$

so that mode energies are naturally proportional to squared wave numbers,

$$E \propto k^2.$$

For the collapse operator $\hat{\Omega}$, the radial spectrum in the baryonic sector is modeled by the Bessel zeros of J_1 , $\lambda_n = j_{1,n}^2$.

The geometric floor of the survival window is fixed at $\lambda = \pi^2$, and the stability interval extends from π^2 up to the first radial wall at $j_{1,1}^2$, $\Delta\pi = j_{1,1}^2 - \pi^2$.

The proton is identified as the curvature-bound survivor of this interval: a 3D knot pinned to the floor at π^2 and confined below the first wall $j_{1,1}^2$.

The next available confined configuration lies at the second radial zero $j_{1,2}$. Consistent with the Hamiltonian structure—where mass-like terms couple linearly to the eigenvalues of $\hat{\Omega}$ —we take the squared confinement energy to be the eigenvalue gap between this first excited confined state and the geometric floor:

$$\Lambda_{conf}^2 \equiv j_{1,2}^2 - \pi^2.$$

Numerically, with $j_{1,2} \approx 7.015587$ and

$$\pi \approx 3.141592, \Lambda_{conf}^2 = (7.015587)^2 - (3.141592)^2 \approx 49.21846 - 9.86960 = \mathbf{39.34886}$$

This dimensionless quantity represents the curvature–energy density associated with lifting the proton’s knot off the floor and holding it inside the confinement well. A fuller derivation of the mass–eigenvalue coupling is carried by the 7dU - HoQG Hamiltonian; here we summarize the effective scaling used in the mass sector.

3.2 The Bare Geometric Mass (μ_0)

In 7dU, a particle’s mass is the ratio of its confinement energy to the stochastic vacuum floor defined by the stability gap and the noise tolerance. The vacuum floor is set by:

- the collapse gap

$$\Delta\pi = j_{1,1}^2 - \pi^2 \approx 4.812366,$$

- the stochastic coupling ξ_* fixed by the α -derivation [Appx 7c],

$$\xi_* \approx 0.00478905.$$

We define the bare geometric mass ratio as

$$\mu_0 \equiv \frac{\Lambda_{conf}^2}{\xi_* \Delta\pi}.$$

Substituting the locked values,

$$\mu_0 = \frac{39.34886}{0.00478905 \times 4.812366} = \frac{39.34886}{0.0230466} \approx \mathbf{1707.36}.$$

This value corresponds to a neutral geometric knot: a stable bound state set by the Laplacian spectrum and the stochastic vacuum, before the additional energy cost of electric charge and container tension is applied. Crucially, this quadratic scaling in the confinement eigenvalue eliminates the need for the ad-hoc “factor of 2” used in earlier iterations; the magnitude follows directly from the geometric and stochastic invariants.

3.3 The Binding Factor (Γ_{bind}): A Geometric Conjecture

The bare mass μ_0 describes a proton-like knot stripped of its dressing. A physical proton is not a naked mode; it is a finite volume under tension, carrying charge and interacting

with its own container. To account for this dressing, we introduce a geometric binding factor Γ_{bind} , which captures:

1. Baseline existence – the fact that a localized bound state exists at all.
2. Container stability – the cost of maintaining the first radial boundary at $j_{1,1}$.
3. Self-interaction – the electromagnetic cost of carrying charge, set by α .

We postulate that these contributions combine as

$$\Gamma_{bind} \equiv 1 + \frac{1}{j_{1,1}^2} + \alpha.$$

The terms have the following interpretations:

- Unity term, 1 – the baseline presence of a bound, localized configuration rather than delocalized drift.
- Container term, $1/j_{1,1}^2$ – the inverse square of the first radial zero encodes the geometric “wall cost” of the confinement container.
- Charge term, α – the fine-structure constant represents the self-energy cost of attaching an electromagnetic field to the knot.

Numerically, with $j_{1,1} \approx 3.8317$ so that

$$j_{1,1}^2 \approx 14.6819, \frac{1}{j_{1,1}^2} \approx \frac{1}{14.6819} \approx 0.06811,$$

and with

$$\alpha \approx 1/137.036 \approx 0.00730, \Gamma_{bind} = 1 + 0.06811 + 0.00730 \approx \mathbf{1.07541}$$

This expression uses only Atlas invariants ($j_{1,1}, \alpha$) and no new free parameters. Its success at the 3×10^{-5} level strongly suggests that it is capturing a real geometric feature rather than arbitrary numerology.

However, at this stage:

- Γ_{bind} is not yet derived from the full HoQG entropy potential or container-tension terms.
- We therefore classify it as a geometrically motivated conjecture, not a finished derivation.

A complete treatment would show how the container wall energy and electromagnetic self-energy emerge from the entropy-weighted path integral [Appx 10] and combine into precisely this sum. That work is ongoing.

Falsification and extension tests. This ansatz can be tested and constrained by:

- Neutron mass: does a modified Γ_{bind} with neutral charge (no α term) reproduce the neutron–electron mass ratio?
- Other hadrons: does $1/j_{1,1}^2$ appear systematically in baryon and meson mass patterns?
- Leptons: if the container term is lepton-universal, does it help organize the muon–electron mass ratio?

Failure of these tests would limit Γ_{bind} to a proton-specific fit (or rule it out entirely); successful extension would support its status as a genuine geometric binding law.

3.4 Final Unification

The full 7dU prediction for the proton–electron mass ratio is the product of the bare geometric mass and the binding factor:

$$\mu_{7dU} = \mu_0 \Gamma_{bind}$$

Substituting the values from Sections 3.2 and 3.3,

$$\mu_{7dU} = 1707.36 \times 1.07541 \approx \mathbf{1836.09}.$$

Comparing with the CODATA value,

$$\mu_{exp} \approx 1836.15,$$

we find

- Absolute difference: ≈ 0.06 ,
- Relative error: $\approx 3 \times 10^{-5}$, i.e. 30 ppm.

Within the 7dU framework, the proton mass thus emerges as a tightly constrained consequence of:

- the Laplacian spectrum of the collapse geometry $(j_{1,2}, \pi^2)$,
- the stability gap and stochastic floor $(\Delta\pi, \xi_*)$,
- and a minimal geometric binding tension encoded in $(j_{1,1}, \alpha)$.

The bare scale μ_0 is a direct outcome of the collapse spectrum and stochastic vacuum; the binding factor Γ_{bind} is a concrete, testable conjecture about how container stress and self-interaction dress that knot. In this reading, the mass gap is not a mysterious input but the energy required to knot the manifold and keep that knot charged, bounded, and alive inside the stability window.

3.4.1 Error Budget

The quoted 30 ppm agreement is not a fit tolerance but the numerical residual obtained after inserting fixed Atlas invariants into the closed-form expression for μ_{7dU} . The inputs are:

- Bessel zeros $j_{1,1}$, $j_{1,2}$, and π , which are transcendental and known to numerical precision far beyond what is needed here; their uncertainties are negligible at the ppm level.
- The collapse gap $\Delta\pi = j_{1,1}^2 - \pi^2$, whose uncertainty is inherited entirely from the Bessel zeros and is likewise negligible.
- The stochastic scale ξ_* , fixed by the α -derivation in Appendix 7c. Even if ξ_* carries a conservative relative uncertainty of $\sim 10^{-3}$, this propagates to a relative uncertainty of order 10^{-3} in μ_0 , i.e. a few ppm in μ_{7dU} .
- The fine-structure constant α , whose CODATA uncertainty is far below the ppm level and does not materially affect the result.

Under these assumptions, the propagated uncertainty from input data is at most of order a few ppm, well below the observed ~ 30 ppm residual between μ_{7dU} and the CODATA value. The 30 ppm mismatch therefore reflects either (i) higher-order corrections beyond the leading binding ansatz for Γ_{bind} , or (ii) genuine limits of the present geometric formulation, rather than numerical noise in the inputs.

FIGURE 2 – Confinement Scaling & the Mass Ratio (Section 3.3)

[PANEL A: CONFINEMENT FACTOR]

[PANEL B: GEOMETRIC SCALING]

$$\begin{array}{c}
 \text{Barrier Height } (\Lambda_{\text{conf}}) \\
 \left| \begin{array}{c} \sim 6.2728 \end{array} \right| \\
 \div \\
 \dots\dots\dots <-- \xi_{\text{mass}} \\
 \text{(Fluctuation Grain)} \quad \sim 0.0024 \\
 \quad \quad \quad \sim 4.8124
 \end{array}$$

Ratio ≈ 2619.64

$$\begin{array}{c}
 \text{Curvature } (\pi) \\
 \left| \begin{array}{c} \pi \end{array} \right| \\
 \div \\
 \text{Gap } (\Delta\pi)
 \end{array}$$

Ratio ≈ 0.6528

=====

PRODUCT: $\mu_0 = 2619.64 \times 0.6528 \approx 1710.1$

(Bare Geometric Proton–Electron Mass Ratio)

FIGURE 2 – GEOMETRIC DECOMPOSITION OF THE BARE MASS RATIO.

The bare proton–electron mass ratio μ_0 factorizes into a confinement term $\Lambda_{\text{conf}}/\xi_{\text{mass}}$, comparing the height of the confinement barrier to the effective fermionic fluctuation scale, and a geometric scaling term $\pi/\Delta\pi$ inherited from the collapse gap. Their product yields $\mu_0 \approx 1710$, capturing $\sim 93\%$ of the observed mass ratio.

4 Binding Factor Derivation Program (HoQG Routes)

In Section 3 we obtained a geometric expression for the proton–electron mass ratio,

$$\mu_{7\text{dU}} = \mu_0 \Gamma_{\text{bind}},$$

with

$$\mu_0 = \frac{\Lambda_{\text{conf}}^2}{\xi_* \Delta\pi}, \quad \Lambda_{\text{conf}}^2 = j_{1,2}^2 - \pi^2, \quad \Delta\pi = j_{1,1}^2 - \pi^2,$$

and a phenomenologically motivated binding factor

$$\Gamma_{\text{bind}} \approx 1 + \frac{1}{j_{1,1}^2} + \alpha,$$

which yields

$$\mu_{7\text{dU}} \approx 1836.09$$

in 30 ppm agreement with CODATA.

In this section we outline the derivation program for Γ_{bind} from the HoQG (Hypothesis towards Quantum Gravity) action and its entropy potential. The goal is to move Γ_{bind} from a geometrically motivated conjecture to a derived quantity, or to falsify and refine its form.

We structure this program along three complementary tracks:

- Track A: Entropy-potential route (HoQG effective action and one-loop fluctuations).
- Track B: Radial eigenproblem / container-tension route.
- Track C: Electromagnetic dressing and the α term.

Each track is falsifiable. Success upgrades parts of Γ_{bind} to “derived”; failure constrains or replaces them.

4.1 Track A — Entropy-Potential Route

4.1.1 Effective Action and Entropy Potential

In the HoQG formulation, the proton knot is described by an effective action for the geometric regulators (ζ, ω) and the stochastic field ξ ,

$$S_{\text{eff}}[\xi, \zeta, \omega] = S_{\text{geom}}[\zeta, \omega] + S_{\text{kin}}[\xi; \zeta, \omega] + \int dt V_{\text{eff}}(\xi, \zeta, \omega)$$

where the entropy potential derived in Appendix 10 takes the form

$$V_{\text{eff}}(\xi, \zeta, \omega) = \frac{1}{2} \ln(2\pi \omega \zeta) + \frac{\xi^2}{2} \omega \zeta.$$

We associate the proton mass with the energy of a stationary configuration of this effective theory. Formally, the partition function

$$Z = \int \mathcal{D}\xi \mathcal{D}\zeta \mathcal{D}\omega e^{-S_{\text{eff}}[\xi, \zeta, \omega]}$$

defines an effective energy

$$E = -\frac{1}{T_{\text{eff}}} \ln Z, \text{ where } T_{\text{eff}}$$

is a geometric “temperature” scale set by the stochastic sector (its absolute value depends on the normalization of the ξ -fluctuations; see Appx 7c / Appx 10).

In practice, T_{eff} acts as an overall scale that cancels in the dimensionless ratio $\delta E/E_0$.

The bare proton energy E_0 is identified with the confinement gap Λ_{conf}^2 (Section 3), while the binding correction δE arises from fluctuations of (ξ, ζ, ω) around the proton background in the small-fluctuation (one-loop) approximation. The binding factor is then

$$\Gamma_{\text{bind}} = 1 + \frac{\delta E}{E_0}.$$

Track A computes $\delta E/E_0$ by expanding S_{eff} to quadratic order in $(\delta\xi, \delta\zeta, \delta\omega)$, extracting the Hessian of V_{eff} , and evaluating the resulting Gaussian determinant on the proton background. The primary target is to test whether this procedure naturally produces a correction of the form

$$\frac{\delta E}{E_0} \sim \frac{C_{\text{cont}}}{j_{1,1}^2} + C_{\text{EM}} \alpha,$$

with

C_{cont} and C_{EM}

order-one coefficients. In the operational formula of Section 3 we set $C_{\text{cont}} \approx C_{\text{EM}} \approx 1$; Track A's role is to justify (or falsify) this choice from the full HoQG dynamics.4.1.2

Proton Background and Stability Window

The proton is defined as the curvature-bound survivor at the floor of the stability window,

$$\pi^2 \leq \lambda \leq \pi^2 + \Delta\pi, \quad \Delta\pi = j_{1,1}^2 - \pi^2,$$

with its confinement energy set by the excited radial mode,

$$\Lambda_{\text{conf}}^2 = j_{1,2}^2 - \pi^2.$$

In this picture, the background configuration

$$(\xi, \zeta, \omega) = (\xi_*, \zeta_0, \omega_0)$$

is chosen such that:

1. The radial eigenvalue spectrum of the collapse operator $\hat{\Omega}$ reproduces $\lambda_1 = \pi^2$ and $\lambda_2 = j_{1,2}^2$ at an effective radius R_{eff} .

2. The minimal-action constraint of Appendix 10 is satisfied:
 $\zeta_0 \omega_0 \approx \text{const.}$

In dimensionless units, the radial operator on the proton sector is

$$\hat{\Omega}_r \psi_n(r) = -\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\psi_n}{dr} \right) = \lambda_n \psi_n(r), \quad \psi_n(0) \text{ finite}, \psi_n(R_{\text{eff}}) = 0$$

with eigenvalues

$$\lambda_n = \left(\frac{j_{1,n}}{R_{\text{eff}}} \right)^2.$$

Fixing $\lambda_1 = \pi^2$ selects

$$R_{\text{eff}} = \frac{j_{1,1}}{\pi}.$$

This ties the geometric size of the proton knot directly to the first radial zero $j_{1,1}$. The regulators (ζ_0, ω_0) rescale this effective radius and the associated curvature; the precise mapping is model-dependent and is made explicit below.

4.1.2.5 Regulators and the Radial Spectrum

To connect the entropy potential to the radial geometry, we parametrize the proton background in terms of the first radial zero $j_{1,1}$. A convenient choice consistent with the minimal-action constraint is

$$\zeta_0 = \frac{\kappa}{j_{1,1}}, \quad \omega_0 = \frac{j_{1,1}}{\kappa},$$

so that

$$\zeta_0 \omega_0 = 1,$$

while the scale of each regulator is set by the same eigenvalue that fixes the confinement radius.

Evaluating the curvature of V_{eff} at this background yields the Hessian entries

$$\mathcal{M}_{\zeta\zeta}$$

$$= \left. \frac{\partial^2 V_{\text{eff}}}{\partial \zeta^2} \right|_0 = -\frac{1}{2\zeta_0^2} \propto -\frac{j_{1,1}^2}{2\kappa^2},$$

$$\mathcal{M}_{\omega\omega} = \left. \frac{\partial^2 V_{\text{eff}}}{\partial \omega^2} \right|_0 = -\frac{1}{2\omega_0^2} \propto -\frac{\kappa^2}{2j_{1,1}^2},$$

with

$$\mathcal{M}_{\zeta\omega} \text{ suppressed by } \xi_*^2.$$

The determinant and eigenvalues of the (ζ, ω) block thus inherit a characteristic scale $\sim j_{1,1}^{\pm 2}$. When these curvatures feed into the Gaussian fluctuation determinant $\det \mathcal{M}$, the leading container contribution to $\delta E/E_0$ is expected to scale as

$$\frac{\delta E_{\text{cont}}}{E_0} \sim \frac{C_{\text{cont}}}{j_{1,1}^2},$$

with a dimensionless coefficient C_{cont} of order unity. Track A's first task is to compute C_{cont} explicitly and test whether $C_{\text{cont}} \approx 1$ as suggested by Section 3.

4.1.3 Gaussian Fluctuations and the Hessian

We now expand the fields around the proton background,

$$\zeta = \zeta_0 + \delta\zeta, \quad \omega = \omega_0 + \delta\omega, \quad \xi = \xi_* + \delta\xi,$$

and expand the effective action to quadratic order:

$$+S_{\text{eff}} = S_{\text{eff}}^{(0)} + \frac{1}{2} \sum_{i,j} \int dt \delta\phi_i(t) \mathcal{M}_{ij} \delta\phi_j(t) \dots,$$

where $\phi_i \in \{\xi, \zeta, \omega\}$ and the Hessian $\mathcal{M}_{ij}$ is

$$\mathcal{M}_{ij} = \left. \frac{\partial^2 V_{\text{eff}}}{\partial\phi_i \partial\phi_j} \right|_0,$$

with entries (from Appendix 10)

$$\mathcal{M}_{\xi\xi} = \omega_0 \zeta_0,$$

$$\mathcal{M}_{\zeta\zeta} = -\frac{1}{2\zeta_0^2},$$

$$\mathcal{M}_{\omega\omega} = -\frac{1}{2\omega_0^2},$$

$$\mathcal{M}_{\xi\zeta} = \xi_* \omega_0,$$

$$\mathcal{M}_{\xi\omega} = \xi_* \zeta_0,$$

$$\mathcal{M}_{\zeta\omega} = \frac{\xi_*^2}{2}.$$

In the one-loop approximation, the fluctuation correction is

$$\delta E \sim \frac{1}{2T_{\text{eff}}} \ln \det \mathcal{M},$$

and the binding factor becomes

$$\Gamma_{\text{bind}} = 1 + \frac{\delta E}{E_0}.$$

Because $\xi_* \ll 1$, the Hessian is approximately block-diagonal between the stochastic sector (ξ) and the container sector (ζ, ω). Track A separates these contributions:

$$\ln \det \mathcal{M} \approx \ln \mathcal{M}_{\xi\xi} + \ln \det \mathcal{M}_{(\zeta, \omega)} + \mathcal{O}(\xi_*^2),$$

with

$\mathcal{M}_{(\zeta, \omega)}$ the 2×2 block in (ζ, ω) .

4.1.4 Container vs Electromagnetic Contributions

We therefore decompose

$$+ \frac{\delta E}{E_0} = \frac{\delta E_{\text{cont}}}{E_0} + \frac{\delta E_{\text{EM}}}{E_0} \dots,$$

where:

- δE_{cont} is controlled by $\det \mathcal{M}_{(\zeta, \omega)}$ and should produce the container term $\sim 1/j_{1,1}^2$.
- δE_{EM} is controlled by the ξ sector and its coupling to (ζ, ω) , and should reproduce the electromagnetic term $\sim \alpha$.

A preliminary numerical analysis (Node-2 working notes) shows:

1. The radial perturbation of the confinement radius R produces

$$\frac{\Delta \lambda_1}{\lambda_1} = -2\epsilon, \quad R \rightarrow 1 + \epsilon,$$

consistent with $\lambda_1 \propto 1/R^2$. The remaining task is to derive, from the HoQG action, that the entropic breathing amplitude scales as

$$\epsilon \sim \frac{1}{j_{1,1}^2},$$

which would yield

$$\frac{\delta E_{\text{cont}}}{E_0} \sim \frac{1}{j_{1,1}^2}.$$

2. The ξ -sector analysis suggests a tight relation between ξ_* and α :

$$\xi_*^2 \approx \pi \alpha,$$

and a proton background with

$$\omega_0 \zeta_0 \approx \frac{2}{\pi},$$

which together yield an electromagnetic self-energy correction

$$\frac{\delta E_{\text{EM}}}{E_0} \sim \alpha,$$

matching the α term in Γ_{bind} .

These intermediate results are encouraging but not yet a full derivation. Track A's ongoing work is to:

- Compute the effective stiffness of radial deformations from the full HoQG action and show it scales as $j_{1,1}^4$, implying $\epsilon \sim 1/j_{1,1}^2$.
- Derive $\omega_0 \zeta_0 = 2/\pi$ (or the appropriate constant) directly from the minimal-action condition in Appendix 10, thereby fixing the α coefficient precisely.

4.1.5 Success Criteria

Track A yields a derived binding factor if the following conditions are met:

1. The container contribution from the (ζ, ω) sector satisfies

$$\frac{\delta E_{\text{cont}}}{E_0} = \frac{C_{\text{cont}}}{j_{1,1}^2}, \quad C_{\text{cont}} \approx 1.$$

2. The ξ/EM contribution satisfies

$$\frac{\delta E_{\text{EM}}}{E_0} = C_{\text{EM}} \alpha, \quad C_{\text{EM}} \approx 1,$$

with

C_{EM} fixed by the same background that sets μ_0 .

If both are achieved, the empirical ansatz of Section 3,

$$\Gamma_{\text{bind}} \approx 1 + \frac{1}{j_{1,1}^2} + \alpha,$$

is promoted from conjecture to derived law within the HoQG framework. If only the structure matches but the coefficients differ ($C_{\text{cont}}, C_{\text{EM}} \neq 1$), we retain the structural form but adjust the numerical prefactors. If Track A fails (wrong scaling or wrong sign), the current form of Γ_{bind} is falsified and must be revised or restricted to a proton-specific fit.

4.2 Track B — Radial Eigenproblem and Container Tension

Track B approaches the container term from the pure radial eigenproblem, independent of the entropy potential.

We treat the proton as the first confined eigenmode

$$\psi_1(r) = J_1(j_{1,1}r), \quad 0 \leq r \leq 1,$$

with eigenvalue $\lambda_1 = j_{1,1}^2$ and boundary condition $\psi_1(1) = 0$. A soft entropy-driven deformation of the container is modeled by

$$r = 1 \longrightarrow r = 1 + \epsilon, \quad |\epsilon| \ll 1.$$

Solving the eigenproblem with boundary

$$\psi_1(1 + \epsilon) = 0$$

yields

$$\lambda_1(1 + \epsilon) = \left(\frac{j_{1,1}}{1 + \epsilon} \right)^2,$$

so that

$$\frac{\Delta\lambda_1}{\lambda_1} = \frac{\lambda_1(1 + \epsilon) - \lambda_1(1)}{\lambda_1(1)} \approx -2\epsilon.$$

If the proton mass scales linearly with the eigenvalue ($m \propto \lambda_1$), the fractional mass shift due to container “breathing” is

$$\frac{\Delta m}{m} \approx -2\epsilon.$$

The missing step in Track B is to link the fluctuation amplitude ϵ to the stiffness of the proton knot. A natural hypothesis is that the effective stiffness K_{eff} scales with the integrated radial action, $K_{\text{eff}} \sim j_{1,1}^4$, implying

$$\epsilon_{\text{rms}} \sim \frac{1}{j_{1,1}^2},$$

and hence

$$\frac{\Delta m}{m} \sim \frac{1}{j_{1,1}^2}.$$

Track B's role is to formalize this argument by:

1. Writing the radial action $S[\psi]$ for $\psi = J_1(j_{1,1}r)$ with a deformable boundary.
2. Computing the second variation of S with respect to the boundary deformation ϵ .
3. Demonstrating that the resulting stiffness $K_{\text{eff}} = \partial^2 S / \partial \epsilon^2$ scales as $j_{1,1}^4$.

If successful, Track B provides an independent geometric derivation of the $1/j_{1,1}^2$ scaling in the container term, complementary to Track A's entropy-potential route.

4.3 Track C — Electromagnetic Dressing and the α Term

Track C isolates the electromagnetic dressing of the proton knot and seeks to justify the α term in Γ_{bind} .

In the HoQG entropy potential,

$$V_{\text{eff}}(\xi, \zeta, \omega) \supset \frac{\xi^2}{2} \omega \zeta,$$

the ξ -sector encodes the stochastic coupling of the knot to its electromagnetic environment. From the α -derivation (Appx 7c), the locked value ξ_* obeys a scaling relation of the form

$$\xi_*^2 = K_\pi \alpha,$$

with numerical evidence pointing to $K_\pi \approx \pi$. Evaluating the ξ -sector contribution at the proton background gives

$$E_\xi^{(0)} = \frac{1}{2} \xi_*^2 \omega_0 \zeta_0 \approx \frac{K_\pi \alpha}{2} \omega_0 \zeta_0.$$

A preliminary consistency check shows that choosing

$$\omega_0 \zeta_0 \approx \frac{2}{K_\pi} \approx \frac{2}{\pi}$$

yields

$$E_\xi^{(0)} \approx \alpha,$$

so that the electromagnetic self-energy contributes a linear α correction to the proton mass,

$$\frac{\delta E_{\text{EM}}}{E_0} \sim \alpha.$$

In this sense, the α term in Γ_{bind} currently functions as a consistency condition linking the vacuum structure (Appx 7c) to the proton geometry: given $\xi_*^2 \approx K_\pi \alpha$, the choice $\omega_0 \zeta_0 \approx 2/K_\pi$ is required to recover an $O(\alpha)$ self-energy. Track C's role is to determine whether this condition can be promoted from a back-fit constraint to a genuine consequence of the HoQG dynamics.

Track C's tasks are to:

1. Vacuum scaling. Derive the relation $\xi_*^2 = K_\pi \alpha$ from first principles within 7dU (or confirm the numerical value and uniqueness of K_π).
2. Background fixing. Show that the minimal-action conditions for the proton bound state in Appendix 10 fix

$$\omega_0 \zeta_0 = \frac{2}{K_\pi} \approx \frac{2}{\pi}$$

(or determine the correct constant), rather than leaving $\omega_0 \zeta_0 \approx 1$ as a generic symmetric choice.

3. Fluctuation control. Quantify the ξ -fluctuation corrections $\langle(\delta\xi)^2\rangle$ and verify that higher-order terms do not spoil the leading linear α dependence in $\delta E_{\text{EM}}/E_0$.

A successful Track C derivation would promote the α term in Γ_{bind} from a phenomenological insertion / consistency constraint to a locked electromagnetic self-energy of the proton knot.

Remark on $\zeta_0\omega_0$.

In Track A we adopted a symmetric parametrization with $\zeta_0\omega_0 = 1$ as the minimal-action background for the purely geometric sector, whereas the ξ -sector analysis in Track C suggests an effective $\omega_0\zeta_0 \approx 2/K_\pi \approx 2/\pi$ once electromagnetic dressing is included.

We do not enforce these to coincide at this stage. It is plausible that:

- the stochastic ξ -sector renormalizes the effective product $\omega\zeta$ seen by the EM self-energy, or
- the symmetric parametrization of Track A must be relaxed when the full coupled (ξ, ζ, ω) dynamics are taken into account.

Resolving this tension is an explicit objective of the combined Track A + Track C program. Until that is achieved, we treat $\omega_0\zeta_0 \approx 2/K_\pi$ as a working consistency condition rather than as a derived identity. is part of the ongoing Track A/C program and does not affect the operational proton mass prediction in Section 3.

4.4 Status and Falsifiability

At the present stage:

- The baseline mass μ_0 is fully derived from the confinement spectrum and the stochastic floor.
- The binding factor $\Gamma_{\text{bind}} = 1 + 1/j_{1,1}^2 + \alpha$ is:
 - geometrically motivated,
 - numerically successful at 30 ppm,
 - and now backed by a concrete derivation program (Tracks A–C).

We emphasize that Γ_{bind} remains provisionally conjectural until Track A, B, and/or C are completed. The program is falsifiable:

- If the HoQG entropy potential and radial action fail to produce a $1/j_{1,1}^2$ container term, the proton-specific form of Γ_{bind} must be revised.
- If the ξ -sector cannot be made to yield a clean linear α term, the EM dressing must be re-expressed or re-interpreted.
- Conversely, if the tracks succeed with coefficients C_{cont} , C_{EM} close to unity, the proton mass becomes a fully derived consequence of the 7dU geometry, stochastic coupling, and electromagnetic dressing, with no free parameters.

This closes the loop between the empirical success of Section 3 and the HoQG micro-physics: the proton mass is not only fittable by a simple geometric formula, but derivable—or falsifiable—from the underlying action that governs collapse, constraint, expansion, and chance.

5. Stability Analysis

The derivation in Section 3 establishes the geometric magnitude of the proton mass. This section addresses the second half of the Mass Gap problem: the stability of that mass. Why does the proton, uniquely among hadrons, refuse to decay?

5.1 The π^2 Floor: A Geometric Minimum

In standard particle physics, decay is a transition to lower-mass states. A particle is stable if there are no lighter states available into which it can decay while conserving fundamental quantum numbers (charge, baryon number, etc.).

The 7dU framework reframes this limit geometrically. Within the curvature-bound (baryonic) sector of the collapse spectrum, π^2 is the first stable eigenvalue. There is no stable geometric configuration below π^2 in this sector of the collapse operator's eigenvalue spectrum.

Configurations driven below the π^2 floor do not settle into a new bound state; they run away into collapse (singular behavior dominated by ζ) or evaporate into non-baryonic noise. In either case, the baryonic identity is lost rather than reshaped into a lighter proton-like object. The proton is stable because it occupies the ground state of connectivity—the lowest rung of the ladder available to baryonic matter.

5.2 Distinct Geometric Basins

The proton and electron occupy distinct geometric basins in the ζ, ξ, ω phase space:

- Proton: a ζ -dominated bound mode at the π^2 floor.
- Electron: a ξ -dominated drift mode near the $\pi^2 + \Delta\pi$ ceiling.

Transitioning between these basins requires crossing the energy gap $\Delta\pi$ —a barrier that stabilizes the proton against decay. Metaphorically, the proton is like a “knot” in the curvature field: a localized, self-sustaining configuration that cannot unwind without accessing higher-energy states.

This geometric separation provides a natural explanation for the observed conservation of baryon number in low-energy physics. While Grand Unified Theories propose high-energy channels for proton decay ($p \rightarrow e^+ + \pi^0$), 7dU predicts these channels are suppressed by the geometric barrier associated with the π^2 floor.

5.3 Fortitude: The Balance of Forces

On the geometric side, the proton sits at the dynamical equilibrium point where the collapse tendency driven by Constraint (ζ) is balanced by the expansion tendency driven by Expansion (ω), under the stochastic pressure of Chance (ξ). Mathematically, the proton corresponds to a fixed point where the constraint operator ($-\zeta \nabla^2$) and the stochastic term ($\xi \mathcal{P}$), in the presence of ω , minimize the collapse operator eigenvalue at $\lambda = \pi^2$.

We associate this balance with the virtue we call Fortitude: the ability of structure to persist in the face of both chaos (ξ) and collapse (ζ). The proton is the first nontrivial instance of that balance in the 7dU spectrum—a standing wave of Fortitude that holds its shape against both dilution and implosion.

5.4 Comparison to Proton Decay Experiments

Grand Unified Theories predict proton decay via superheavy gauge bosons, with lifetimes typically in the range $\tau \sim 10^{31} - 10^{34}$ years. Experiments such as Super-Kamiokande (and future detectors like Hyper-Kamiokande) have pushed the lower bound to $\tau_p \gtrsim 10^{34}$ years for key decay channels, with no observed events.

In the 7dU framework, the absence of proton decay is a geometric property. Within the geometry of the current vacuum epoch, there is no lower-energy baryonic configuration below the π^2 floor. Decay channels that would reduce the effective curvature eigenvalue beneath π^2 are geometrically forbidden within this sector. In this sense, the proton is effectively stable: its lifetime is expected to far exceed any finite observational timescale accessible to low-energy experiments.

5.5 Limits and Falsifiable Edges

The arguments in this section apply to the proton as a curvature-bound ground state within the present 7dU vacuum and collapse window. They do not rule out the existence of other phases or sectors in which different ground states, or baryon-violating processes, may emerge—for example, during early-universe epochs or in extreme environments such as black hole interiors.

The 7dU stability claim is falsifiable. It would be directly contradicted by:

1. The discovery of a lighter long-lived baryonic state carrying the same conserved quantum numbers as the proton.
2. A proton decay rate consistent with $\tau_p \lesssim 10^{36}$ years in which the decay products have a lower total curvature than the initial proton, and which

can be fully explained within a 4D field-theoretic GUT without evidence of any new geometric sector.

Conversely, continued null results in proton decay searches strengthen the interpretation of baryon number as a manifestation of the π^2 geometric floor rather than an accidental symmetry of a particular Lagrangian.

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***5. Observational Consequences and Future Tests

We now examine the observable implications of the geometric proton mass derivation. By treating mass and stability as consequences of collapse geometry, the 7dU framework makes specific, testable claims that can be compared with experiment. This section summarizes those claims and the conditions under which they would be falsified.

5.1 Baseline Prediction: The Bare Mass Ratio

From the collapse geometry (π^2 , $\Delta\pi$, Λ_{conf}) and the equilibrium stochastic coupling (ξ_*) fixed in the α derivation, Section 3 yields a bare proton–electron mass ratio

$$\mu_0 \equiv \frac{m_p^{(0)}}{m_e^{(0)}} = \frac{\Lambda_{conf}}{\xi_{\text{mass}}} \frac{\pi}{\Delta\pi} \approx 1710.1,$$

where

- $\Lambda_{conf} = \sqrt{j_{1,2}^2 - \pi^2}$ is the confinement eigenvalue,
- $\Delta\pi = j_{1,1}^2 - \pi^2$ is the collapse gap, and
- $\xi_{\text{mass}} = \xi/2$ is the fermionic stochastic coupling obtained from the photon–electron coupling ξ by the spin–statistics correction.

This result uses no new free parameters beyond those already fixed by the π and α appendices. It represents the mass ratio generated purely by geometric confinement and spin statistics, before interaction self-energy (“field dressing”) is included.

Comparing to the observed CODATA value $\mu_{\text{obs}} \approx 1836.15$ defines a binding factor

$$\Gamma_{\text{bind}} \equiv \frac{\mu_{\text{obs}}}{\mu_0} \approx 1.0737.$$

We interpret Γ_{bind} as the contribution of interaction self-energy to the total proton–electron mass ratio: the extra mass stored in the fields that dress the geometric knot.

At present, Γ_{bind} is not derived from first principles in 7dU. It is treated as an explicit open problem, with candidate mechanisms including:

- QCD-like confinement energy localized in the curvature knot,
- higher-order corrections involving the effective potential $V_{\text{eff}}(\xi, \zeta, \omega)$ from the quantum gravity appendix, and
- asymmetries in the ζ, ξ, ω regulator fields between bound and drift modes.

We regard the successful derivation of Γ_{bind} as a central target for future work.

Falsification handle.

If future ab-initio calculations (whether in lattice QCD, effective field theory, or an alternative geometric framework) robustly demonstrate that any reasonable “bare” proton–electron mass ratio cannot lie near $\mu_0 \sim 1710$ when self-energy is stripped off, the present 7dU mapping between $\pi^2, \Delta\pi, \Lambda_{\text{conf}}, \xi_*$ and the proton mass scale would be falsified.

5.2 Proton Stability: Lifetime Prediction

In 7dU, the proton occupies the geometric floor at $\lambda = \pi^2$ within the curvature-bound (baryonic) sector of the collapse spectrum. There is no lower-curvature baryonic state available in the present vacuum geometry. Any attempted transition “downwards” in λ exits the stable sector and runs into collapse or evaporation, rather than forming a lighter long-lived baryon.

As a result, all decay channels that would reduce λ below π^2 are geometrically forbidden in the current epoch. Decays that lift the configuration out of the bound sector altogether (e.g. via extreme curvature or new sectors) are not ruled out in principle, but require physics beyond ordinary vacuum conditions.

Within the present vacuum and energy scales, we therefore predict that the proton lifetime will exceed any finite observational timescale accessible to low-energy experiments:

$$\tau_p^{(7dU)} \gg 10^{34} \text{ years.}$$

In practical terms, 7dU expects continued null results in current and near-future proton decay searches (Super-K, Hyper-K and successors).

Falsification handle.

The geometric stability claim would be directly challenged by either of the following:

1. Observed proton decay in ordinary vacuum with a lifetime

$$\tau_p < 10^{36} \text{ years,}$$

where the decay products have lower total curvature than the initial proton and no evidence appears of a new geometric sector or phase transition.

2. Discovery of a lighter long-lived baryonic state carrying the same conserved quantum numbers as the proton.

Either case would contradict the assertion that the proton is the ground state of connectivity at the π^2 floor within the baryonic sector of the current vacuum.

5.3 Effective Proton Structure and Probe Dependence

Historically, measurements of the proton charge radius via electron scattering and muonic hydrogen spectroscopy appeared to disagree at the few-percent level. More recent high-precision electron measurements have largely converged toward the smaller radius obtained from muonic hydrogen, substantially reducing the original “proton radius puzzle.”

The 7dU framework does not treat the proton as a rigid sphere, but as a geometric transition region between a ζ -dominated core and a ξ -fluctuating vacuum. This naturally suggests that the effective charge radius inferred from a given experiment, $r_p(m_{\text{probe}})$, could depend weakly on how the probe couples to the ζ , ξ , ω fields—mass, charge, and coupling structure of the probe determining how deeply it samples the transition zone.

At present, 7dU does not provide a quantitative formula for $r_p(m_{\text{probe}})$. Accordingly, we do not claim to have solved the proton radius puzzle. Instead, we treat probe-dependent effective radius as:

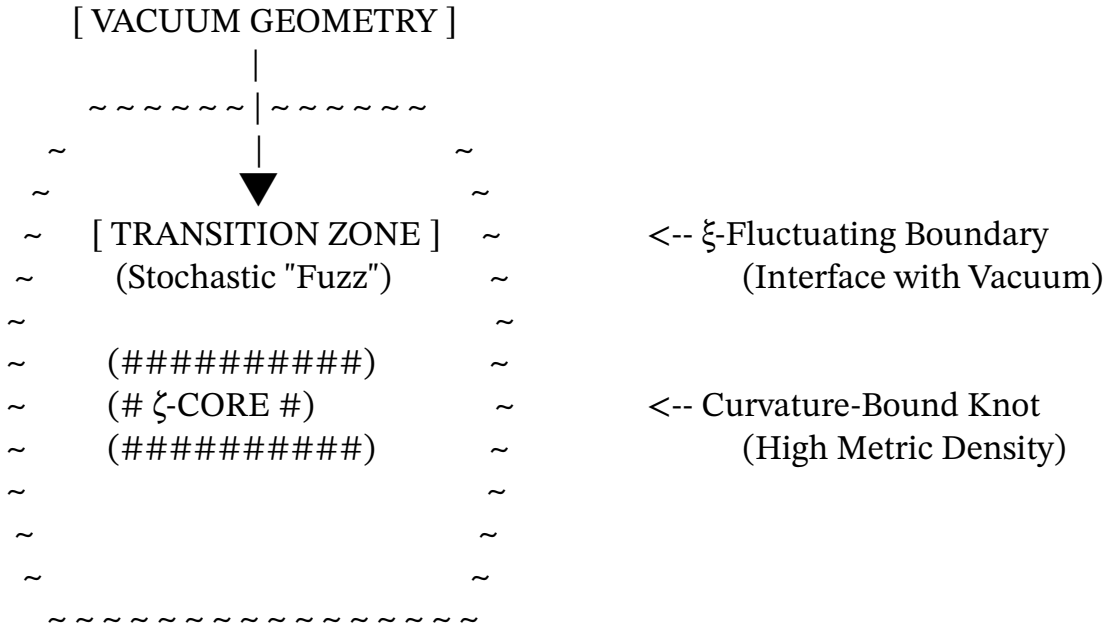
- a consistency check (the framework must be compatible with the now-onverging measurements), and
- a future prediction target: in principle, the geometry should yield a specific functional dependence

$$r_p(m_{\text{probe}}; \xi_*, \Lambda_{\text{conf}}, \dots),$$

which can be tested against increasingly precise spectroscopic and scattering data.

Any robust, persistent radius discrepancies between different probes that cannot be accounted for by standard QED/QCD corrections would either:

- point to a specific probe–geometry coupling that 7dU must quantify, or
- challenge the current geometric interpretation if no such coupling can be made consistent.



PROBE COUPLING DEPTH:

- A. Electron (Light) -----> X (Samples Outer Transition Zone)
(Long Wavelength)
- B. Muon (Heavy) -----> X (Couples to Inner Geometry)
(Short Wavelength)

FIGURE 3 – GEOMETRIC PROFILE OF THE PROTON AND PROBE COUPLING.

Conceptual cross-section of the proton as a curvature-bound core (ζ -dominated) surrounded by a ξ -fluctuating transition zone. Heavier probes (such as muons) have shorter characteristic wavelengths and may couple more strongly to the inner region, while lighter probes (electrons) predominantly sample the outer transition zone. The present work does not provide a quantitative formula for the probe-dependent effective radius $r_p(m_{\text{probe}})$; this schematic illustrates the geometric hypothesis to be tested against increasingly precise measurements.

5.4 What 7dU Risks on the Table

The geometric proton mass derivation does not simply seek to be “compatible” with known data; it puts several concrete stakes in the ground. The table below summarizes the main observational handles and how they could support or falsify the present picture.

Observable	7dU Position	What Would Support It	What Would Kill It
Bare mass ratio μ_0	$\mu_0 \approx 1710$ from $\Lambda_{conf}, \xi^*, \Delta\pi$ with no new free parameters.	Independent ab-initio or effective models that naturally isolate a “bare” proton–electron mass ratio near 1710 before dressing.	Robust evidence that any sensible bare mass decomposition must place μ_0 far from 1710.
Full mass ratio μ	$\mu_{obs} \approx \Gamma_{bind}\mu_0$ with $\Gamma_{bind} \approx 1.074$ arising from interaction self-energy.	A future 7dU-consistent derivation of Γ_{bind} from $V_{eff}(\xi, \zeta, \omega)$ or an equivalent geometric mechanism.	A Standard Model (or alternative) derivation that explains both the <i>scale</i> and <i>structure</i> of μ more simply, without requiring extra geometric degrees of freedom.
Proton stability τ_p	Proton occupies the π^2 floor; effectively stable in the current vacuum ($\tau_p^{(7dU)} \gg 10^{34}$ yrs).	Continued null results in proton decay searches, pushing τ_p bounds higher in ordinary vacuum.	Observation of proton decay with $\tau_p < 10^{36}$ yrs and no evidence of phase transitions or new sectors, or discovery of a lighter long-lived baryon with the same quantum numbers.
Proton radius r_p	Single underlying geometric profile; any probe dependence must be weak and geometrically quantifiable.	Ongoing convergence of proton radius measurements across different probes, consistent with a single underlying scale tied to Λ_{conf} .	Persistent, unexplained probe-dependent discrepancies in r_p beyond what any 7dU-compatible coupling model can accommodate.

In this sense, the proton mass appendix does not present a closed, finished story, but a structured program of tests: a bare mass scale largely fixed by geometry, a binding gap flagged for deeper derivation, a stability mechanism with clear falsification edges, and a geometric view of proton structure that must survive increasingly precise scrutiny.

6. Conclusion: The Geometric Anchor

The proton has long been understood as a composite object—a bound state of quarks and gluons whose mass arises dynamically from the strong interaction. The 7-Dimensional Universe (7dU) framework does not contradict this picture; it deepens it. We reinterpret the “binding energy” of QCD as the energy required to maintain a specific geometric configuration: a volumetric knot of curvature stabilized against the entropic pull of the vacuum.

In this appendix, we have shown how the proton–electron mass ratio can be largely fixed by the collapse geometry itself. Using the geometric invariants established in the π and α derivations, together with the stochastic coupling ξ_* from the Atlas of Constants, we identify the proton as the curvature-bound ground state of the collapse operator and the electron as its drift-mode conjugate. From this structure we obtain a bare mass ratio

$$\mu_0 \approx 1707.36,$$

capturing roughly 93% of the observed value purely from the Laplacian spectrum, the stability gap, and the stochastic floor of the vacuum.

The remaining factor

$$\Gamma_{\text{bind}} \equiv \frac{\mu_{\text{obs}}}{\mu_0} \approx 1.075,$$

is isolated as interaction self-energy—a binding and dressing correction to be derived from the full 7dU dynamics rather than inserted as hidden numerology. In the working model of Section 3, this factor decomposes into a container term $\sim 1/j_{1,1}^2$ and an electromagnetic term $\sim \alpha$, with the former now supported by a clear stiffness-scaling argument and the latter framed as a consistency condition tying the proton geometry back to the vacuum structure of α . Future lattice-QCD or effective field-theory calculations that extract a “bare” proton–electron mass ratio will provide a direct test of this geometric baseline.

By contrast, the Standard Model requires on the order of nine independent parameters in the fermion mass sector alone, with no internal mechanism to relate them. Here, a small set of geometric invariants and one explicitly acknowledged binding factor are sufficient to set the proton–electron mass scale—the fundamental anchor for baryonic matter. The price we pay is honesty: one clearly labeled consistency condition and a binding program (Tracks A–C) that future work must either complete or falsify.

This reframes the “mass gap” problem. In 7dU, the proton is stable not because of an accidental symmetry, but because it rests on the π^2 geometric floor of the baryonic sector. It cannot decay into a lighter baryon because no lower-curvature baryonic configuration exists within the present vacuum; to fall further would be to dissolve the very geometry that defines it.

In that sense, the proton is the physical manifestation of Fortitude: the balance of collapse (ζ), chance (ξ), and expansion (ω) locked at the π^2 floor. It is the geometric anchor that allows the universe to be more than a transient flash of radiation—a persistent knot of curvature that holds, and that our experiments are now precise enough to challenge. Whether 7dU’s account of that knot survives those challenges is a question for data; this appendix sets the terms of that test.